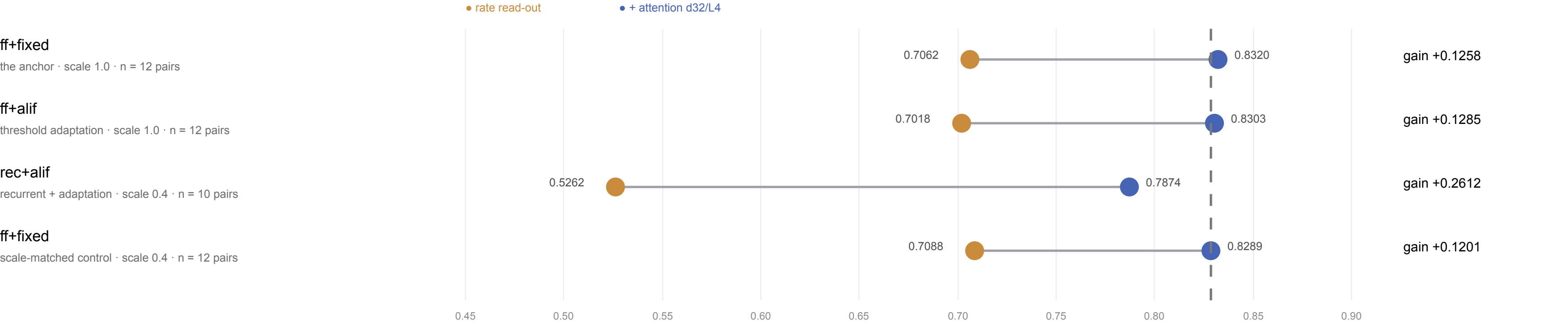


Figure S — Substrate: the read-out does not substitute for temporal state

h128 / published-2ms / adjacent-sum-5 / e400 / d32-L4, seed-paired. The anchor campaign's 720 cells all sat on ff+fixed, so its gain had two readings it could not separate.

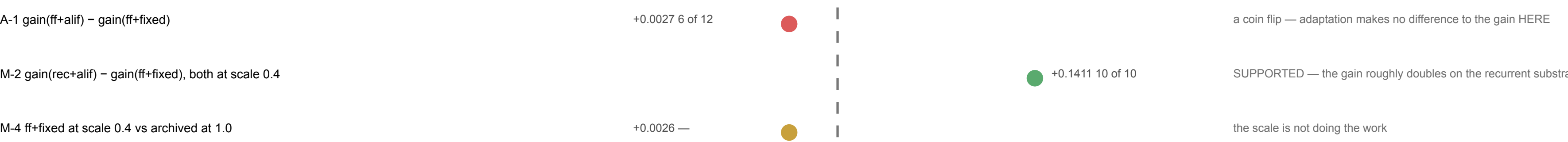
ADDS temporal structure no such substrate represents, or SUBSTITUTES for the adaptation and recurrence ff+fixed happens not to have. ETLP makes the second reading the live one.

Panel A — one shared accuracy axis, in the row order of Table SHD-7. NOT sorted by gain.



THE RECURRENT SUBSTRATE DOES NOT WIN. rec+alif+attn reaches 0.7874 against ff+fixed+attn's 0.8289 at the same scale — attention closes most of the gap the substrate gives away, and not all of it. NO VERDICT IS ISSUED ON THAT ORDERING, and nothing here is sorted by gain.

Panel B — the registered contrasts, each against its two-sided bar



BOTH READINGS OF “THE GAIN DOUBLES”, because neither may be quoted alone. rec+alif starts 0.18 LOWER, with 0.4738 of headroom against 0.2912.

Gain / headroom is 0.551 against 0.412, so the ratio falls 2.2× → 1.34×. The ORDERING SURVIVES; most of its apparent size does not.

That normalisation is POST-HOC AND NOT REGISTERED, and saying so is not licence to prefer whichever of the two numbers is more convenient, in either direction.

Panel C — usability, which had to be measured before any row of Panel A could be. This is why the recurrent row exists at one scale only.

| arm       | scale | completed | voided | diverged |                                                                                                                                                                      |
|-----------|-------|-----------|--------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rec+alif  | 0.4   | 11 / 12   | 0      | 1        | rec+fixed's ten voids are ALL SATURATION — up to 52.3% of hidden units pinned at maximum firing — and NONE is divergence at scale 0.4. So on the recurrent substrate |
| rec+alif  | 1.0   | 8 / 12    | 0      | 4        |                                                                                                                                                                      |
| rec+fixed | 0.4   | 7 / 12    | 5      | 0        | ADAPTATION IS STABILISING, the opposite of the sign that wave's own hypothesis asserted.                                                                             |
| rec+fixed | 1.0   | 5 / 12    | 5      | 2        | A-1 above is therefore inert AT THIS OPERATING POINT, on this substrate — not in general.                                                                            |

TEN PAIRS (10 of 12), THE REGISTERED MINIMUM. The two recurrent arms lost different seeds and one further loss on either would have made M-2 unreportable. Pairing REDUCES survivorship rather than removing it: the surviving pairs are those that did not diverge, and divergence is not random. The feed-forward rows carry no such exposure at 12/12.